RGB-D frames

Perception

Instance Segmentation



MSAM-Net:

Multi-Scale Feature Aggregation

Attention-Guided Mask Optimization

2D Instance Masks

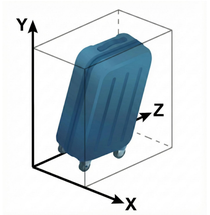
Modeling

Geometric Consistency Filtering



Pixel-to-3D Cross Consistency Conditional Filtering

Temporal Stabilization



Multi-frame Consistency-Aware OBB Stabilization

Stable Refined

Action

State Estimation



Kalman Filter-based Spatiotemporal Estimation

Motion Planning



Two-Stage Planning in Relative Velocity Space

Active State Regulation



Entropy Reduction & State Reset

Execution